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The Rhythmic Interpretation of Monophonic Music

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The assignment of a rhythmic interpretation to a piece of metrical music calls for the postulation of an underlying meter and the parsing of the note values according to this meter. In this article we develop the implications of this view, which include the following propositions.

1. Any given sequence of note values is in principle rhythmically ambiguous, although this ambiguity is seldom apparent to the listener.
2. In choosing a rhythmic interpretation for a given note sequence the listener seems to be guided by a strong assumption: if the sequence can be interpreted as the realization of an unsyncopated passage, then that is how he will interpret it.
3. Phrasing can make an important difference to the rhythmic interpretation that the listener assigns to a given sequence. Phrasing can therefore serve a structural function as well as a purely ornamental one.

Introduction

When a listener hears a tune played or sung without accompaniment, he does not merely perceive the relative pitches and durations of the notes; he becomes aware, to a greater or lesser extent, of the tonal and rhythmic relations between them—for example, whether the tune is in a major or minor key, and whether its rhythm is that of a march or a minuet. In assigning a rhythmic interpretation to a particular passage, the listener will generally be influenced by clues of many different kinds. The notes of the passage may be differently accented, some of them may be played *staccato* and others *legato*, rhythmic clues may be supplied by the tonal relationships between the notes, and in vocal music the words themselves may exert a dominant effect. But there is one source of evidence that is always available, even when there are no words, when the notes are of indefinite pitch, and when the performance is devoid of accent, phrasing or rubato: even in such

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impoverished conditions the listener may still arrive at a rhythmic interpretation of the passage, based solely on the relative durations of the notes. It was, we believe, Simon (1968) who first recognized the need for an account of how listeners perform this lowly perceptual task, and since that time Longuet-Higgins and Steedman (1971), Longuet-Higgins (1976), and Longuet-Higgins and Lee (1982) have suggested partial solutions of the problem. In this article we take a step back and consider the criteria that might lead a listener to favor a particular rhythmic interpretation of a given sequence of notes.

The article falls roughly into three parts. In the second and third sections we develop a generative theory of musical rhythms in the same spirit as the work of Lindblom and Sundberg (1972), but focusing particularly on the rhythms of individual bars and their relationship to the underlying meter. We pay special attention to the concept of syncopation, and in the fourth section—the middle part of the article—we suggest that in assigning a rhythmic interpretation to a sequence of notes the listener tends to avoid interpretations that demand either syncopations within bars or syncopations across bar lines. The fifth section sketches some hypotheses about the perception of higher level rhythmic structure, and in the sixth section—the last main part of the article—we consider the role of the performer in clarifying for the listener what might otherwise be rhythmically ambiguous sequences; in this section we pay special attention to the grouping of notes into phrases by the device of slurring together adjacent notes.

Our data and predictions, like those of Lindblom and Sundberg (1972), refer in the first instance to written music, just as the data and predictions of the theoretical linguist usually refer in the first instance to printed text. The reader may feel that this restriction evades a number of important issues connected with the live performance of real music, but the expressive qualities of live performances vary so widely that we have felt it essential to restrict the discussion in some way, and we have done this by concentrating on those directions about a performance that a composer customarily indicates in a musical score, such as whether a particular note is to be played *staccato* or *legato*. Our appeal is therefore not to the auditory sensibilities of the reader but to that part of his musical intuition that is engaged when he reads a musical score that indicates the values of the notes and the way in which they are meant to be phrased; in a sense we appeal to him as a potential performer, who has to decide how the musical listener is likely to interpret the music if it is performed according to the composer's directions.

We start, then, by considering unadorned sequences of note values and how a listener might assign them rhythmic interpretations. The following examples may convince the reader that the problem is more subtle than it might seem:

(a) An isochronous sequence of notes such as the chimes of a clock has no obvious rhythmic structure, in that the notes might be felt to be grouped in twos or in threes, with or without "upbeats":

But, at least if heard in isolation, the sequence is most unlikely to receive any of the following (theoretically possible) rhythmic interpretations:


(1)


(2)


(3)

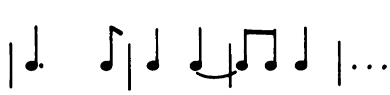
(b) If the relative values of the notes are as shown in (4)


(4)

then the sequence has the obvious rhythmic interpretation (5), rather than (6) or (7):


(5)


(6)


(7)

(c) If, on the other hand, the relative note values are as shown in (8), then the obvious interpretation is (9) rather than, for example, (10) or (11):


(8)

(9)

(10)

(11)

These particular facts (assuming the reader to accept them as such) are, of course, uninteresting in themselves; what is of interest is whether they exemplify any generalization about the perceptual propensities of musicians. We suggest that they do, and in the following paragraphs we attempt to formulate such a generalization as precisely as possible and to show that its implications accord with common musical intuition.

The Theory of Metrical Rhythms

In recent years it has been recognized (Martin, 1972; Lindblom & Sundberg, 1972; Longuet-Higgins, 1976) that there is a close parallel between metrical rhythms (as opposed to the "free" rhythms of plainchant or recitative) and the syntactic structures of sentences. A metrical passage is one that could be assigned a "time signature" such as $\frac{4}{4}$, $\frac{3}{4}$, or $\frac{6}{8}$; what such a signature specifies is actually a grammar consisting of a set of context-free realization rules such as the following:

$\frac{4}{4}$ → 𝅝 or 𝄻 or ($\frac{2}{4}$ + $\frac{2}{4}$)

$\frac{2}{4}$ → 𝅗𝅥 or 𝄼 or ($\frac{1}{4}$ + $\frac{1}{4}$)

$\frac{1}{4}$ → ♩ or 𝄽 or ($\frac{1}{8}$ + $\frac{1}{8}$)

$\frac{1}{8}$ → ♪ or 𝄾 or .....

In such a set of rules the symbols $\frac{4}{4}$, $\frac{1}{8}$, etc. (which have been adopted here for their appeal to musical intuition) are not themselves notes or rests, although they may be realized as such; they correspond to entities such as bars and beats and may be said to designate "metrical units" at various levels in a metrical hierarchy. In the commonest meters each bar or shorter metrical unit is divisible into either two or three metrical units at the next level down; if this is the case the meter may be described as a "standard" meter, and represented by a list consisting entirely of 2's and 3's. Thus the $\frac{4}{4}$ meter specified earlier would be represented by the list [2 2 2 . . .], indicating that the bar—the top-level unit—is divisible into 2, the half-bar is also divisible into 2, and so on.

The time signatures $\frac{3}{4}$ and $\frac{6}{8}$ indicate meters with different realization rules, namely

$\frac{3}{4} \rightarrow$ 𝅗𝅥. or 𝄼· or $(\frac{1}{4} + \frac{1}{4} + \frac{1}{4})$

$\frac{1}{4} \rightarrow$ ♩ or 𝄽 or $(\frac{1}{8} + \frac{1}{8})$

$\frac{1}{8} \rightarrow$ ♪ or 𝄾 or .....

and

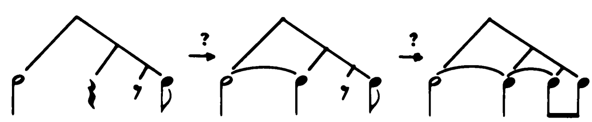
$\frac{6}{8} \rightarrow$ 𝅗𝅥. or 𝄼· or $(\frac{3}{8} + \frac{3}{8})$

$\frac{3}{8} \rightarrow$ ♩. or 𝄽· or $(\frac{1}{8} + \frac{1}{8} + \frac{1}{8})$

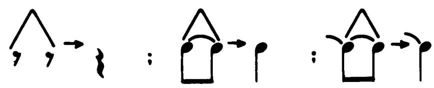
$\frac{1}{8} \rightarrow$ ♪ or 𝄾 or .....

The time signatures commonly used for specifying such meters usually imply only the first one or two divisions of the bar; in the notation that we are proposing the $\frac{3}{4}$ and $\frac{6}{8}$ meters would be symbolized as [3 2] and [2 3] respectively. (We will not enter here into the detailed description of "context-sensitive" meters, in which, for example, the first beat of a bar is divisible into 2 and the second into 3, or of "variable" meters, in which the bar, or any subunit thereof, is optionally divisible into either 2 or 3 units at the next level down.)

By the progressive application of the realization rules of a given meter we generate a "tree" structure, in which the "root" node represents the whole bar, and the other nonterminal nodes represent lower level metrical units. At this point the terminal nodes of the tree are all either (sounded) notes or rests; in order to allow for tied notes (notes that are tied back to their predecessors, rather than being separately sounded), a further rule is now invoked: if any note N, sounded or tied, is immediately followed by a rest R, then R may be replaced by a note that is tied back to N:



Finally, in order to avoid meaningless sequences of rests or tied notes, it is stipulated that any divided metrical unit consisting entirely of rests, or of notes that are tied together, is replaced by an undivided unit composed of a single rest, or of a single note that is sounded or tied according to whether the first note of the replaced unit was sounded or tied:



Any tree that results from the application of a set of metrical realization rules, and the meter-independent replacement rules just stated, is a possible rhythm that accords with the given meter. *The relationship between the rhythm and the meter may be simply stated: The former is one of the structures that is generated by the grammar associated with the latter.* On this view, a rhythm is much more than just a sequence of note values; it is a syntactic structure in which the note values are implicit in the terminal symbols, but do not by themselves define the rhythm; if they did, then no sequence of note values could be rhythmically "ambiguous"—open to alternative rhythmic interpretations—as such sequences undoubtedly are.

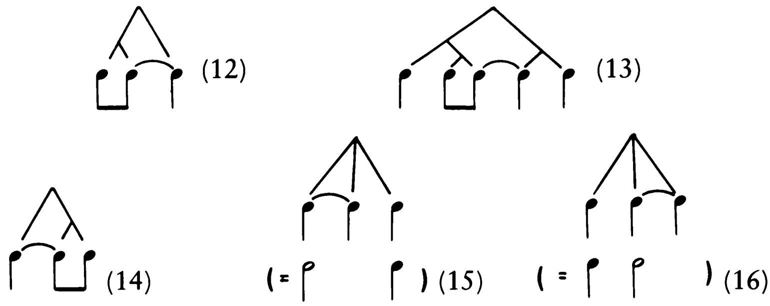
To show that the formal theory just described accounts quite naturally for the phenomenon of rhythmic ambiguity, one example should suffice. A bar consisting of a dotted crotchet (quarter-note) followed by three quavers (eighth-notes) might be, in common parlance, "in either $\frac{3}{4}$ or $\frac{6}{8}$." In the terms of the present theory, such a bar is generated both by the meter [3 2] and by the meter [2 3]:

The relative note values are the same for both, but the structures to which they belong are quite different, and it is the existence of these alternative structures that constitutes the rhythmic ambiguity of the sequence.

This example might suggest that rhythmic ambiguity is a merely sporadic occurrence, like syntactic ambiguity in sentences; but in fact it is an all-pervasive musical phenomenon. As the reader will already have noticed in inspecting examples (a), (b), and (c) of the first section, even the most innocuous sequences of note values permit an unlimited number of rhythmic interpretations. Hence, insofar as listeners agree on the rhythms of given sequences of note values, we can infer that the process of rhythmic perception must be tightly constrained by some tacit assumption as to what counts, or does not count, as a "natural" interpretation. In the following sections we are largely concerned with the uncovering of just such an assumption, but in order to state the assumption precisely, it is first necessary to clarify another rhythmic concept—that of syncopation.

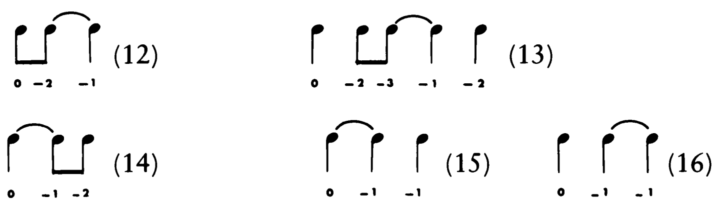
Syncopation

Informally, most musicians would agree that rhythms (12) and (13) are syncopated, whereas (14) and (15) are not; (16) is perhaps a marginal case:



What is the difference? Surely that, in (12) and (13) at least, a "heavier" note is tied back to a "lighter" sounded note, whereas in (14) and (15) the "heavier" note is the first of the tied pair. This observation suggests that in order to pin down the concept of syncopation we need to define the "weight" of a note or a rest; the following definition seems to accord with intuition: *The weight of a given note or rest is the level of the highest metrical unit that it initiates.* (The level of the topmost metrical unit is arbitrarily set equal to 0, and the level of any other unit is assigned the value $n-1$, where n is the level of its "parent" unit in the rhythm.)

On this definition, the weights of the notes in (12) to (16) are as follows:



We are now in a position to define a syncopation:

If R is a rest or a tied note, and N is the next sounded note before R, and the weight of N is no greater than the weight of R, then the pair (N,R) is said to constitute a syncopation. The "strength" of the syncopation is the weight of R minus the weight of N.

Thus in (12) and (13) we have syncopations of strength 1 and 2 respectively, and in (16) a syncopation of strength 0, but (14) and (15) contain no syncopations—as musical intuition demands.

So far our discussion of rhythmic structure has been confined to the individual bar, for which the entire rhythm descends from a single node, the level of this node being arbitrarily set to zero. But many monophonic passages consist of sequences of bars which may or may not be grouped into higher metrical units. The concepts of level and weight may nevertheless be applied to such passages: the level of any metrical unit in a sequence of bars can still be defined as zero if the unit is a full bar and can otherwise be assigned the value that it would have if the bar containing the unit were considered in isolation. The same applies to the concept of weight: the weight of a note or rest can still be defined as the level of the highest metrical unit that it initiates. Then it will be appropriate to describe a sequence of bars as syncopated if and only if one of the bars contains a syncopation, or some bar begins with a rest or a tied note that follows a sounded note of lesser weight.

Regular Passages

At this point the reader is invited to reconsider the note sequences (a), (b), and (c) of the first section. In each case, the "natural" interpretations of the sequence are the only ones that are entirely unsyncopated; in all the alternative interpretations there are notes that are tied back to sounded notes of equal or lesser weight. We may suspect that this is a general characteristic of "natural" interpretations: that when a sequence of notes can be interpreted as the realization of an unsyncopated passage, then the listener will interpret the sequence in this way. But the question immediately arises: If a note sequence is indeed the realization of an unsyncopated passage, how can the listener arrive at such an interpretation of it, using no more information than the relative durations of the notes? This is the problem to which we now turn.

We define a regular passage as a sequence of bars satisfying the following conditions:

1. Every bar, except possibly the first, begins with a sounded note (this ensures that there are no syncopations across bar lines)
2. All the bars are generated by the same standard meter
3. There are no syncopations within any of the bars.

We now show that if a sequence of notes is indeed the realization of a regular passage, then the durations of the sounded notes (defined as the times between their onsets) supply useful evidence about the rhythmic structure. More precisely, we show that if D is the duration of the metrical units at level L, then the sounded notes whose durations exceed D must be spaced by time intervals that are multiples of D′, where D′ is the duration of the units at level L + 1. So by considering notes of progressively greater duration we can, as it were, build up the meter from the bottom, until we reach the lowest metrical unit that accommodates the longest sounded notes. We will now present the steps in the demonstration as a series of lemmas with accompanying proofs.

Lemma 1: Let N be a sounded note, and let U be the highest metrical unit that N initiates. Then the duration of N cannot exceed that of U.

Proof: Let N be any sounded note except the last one in the sequence, and let S be the next sounded note after N; let U be the highest metrical unit that N initiates. Suppose, contrary to the Lemma, that the onset of S occurs later than the end of U. Then the end of U must coincide with the onset of a rest or tied note R; let V be the highest metrical unit initiated by R. Since the beginning of V coincides with the end of U, the level of V must be at least as high as the level of U. Hence the weight of R is no less than the weight of N, and so (N,R) constitutes a syncopation. But by hypothesis the rhythm is unsyncopated, and so the onset of S must occur no later than the end of U, as the lemma asserts.

Lemma 2: If D is the duration of the metrical units at level L, then every sounded note of duration greater than D must be of weight at least L + 1.

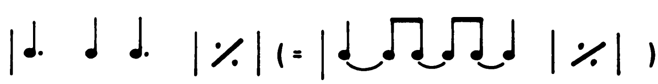
Proof: By Lemma 1, no sounded note of weight L can have a duration greater than D. Therefore any sounded note of duration greater than D must be of weight at least L + 1.

Lemma 3: If L is any level in the meter, and if D is the duration of the units at that level, then the time intervals between the onsets of sounded notes of weight L or greater must be multiples of D.

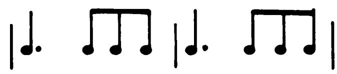
Proof: The duration of any metrical unit of level higher than L must be a multiple of D, because when a metrical unit is divided, its daughter units are of equal duration. But the sounded notes under consideration all initiate metrical units of level L or greater, so the lemma follows.

The task of determining the meter of a regular passage, and thus revealing its rhythmic structure, may therefore be broken down into the following steps:

First one must locate the shortest metrical units, and these will have the same durations as the shortest sounded notes. (This is not true for syncopated passages, such as



where the crotchet does not delimit the shortest metrical unit, which is at the quaver level.) The next step is to consider the set of all sounded notes of duration longer than the shortest metrical unit. By Lemma 3, the times of onset of these notes must be spaced by multiples of the next shortest unit. So if the shortest metrical unit is of duration D, and the highest common divisor of the spacings between sounded notes longer than D is D′, we shall find that D′/D is an integer whose only prime factors are 2 and 3. (This conclusion rests, of course, on the assumption that the meter is standard.) If D′/D turns out to be a power of 2, then we can conclude that the metrical unit at the next level is of duration 2D, and if D′/D is a power of 3, the duration of the unit must be 3D. But if D′/D is a multiple of 6, then the next unit up might have a duration equal to either 2D or 3D; in this case the meter, and consequently the rhythm, will be ambiguous:


…; $\frac{3}{4}$ or $\frac{6}{8}$?

If D′/D has any other value, such as $\frac{1}{2}$, 1, $\frac{3}{2}$, 5, we can infer that there is no interpretation of the sequence as the realization of a regular passage.

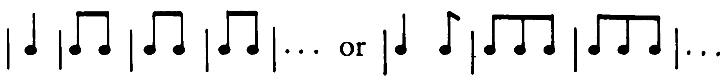
The remaining stages in the determination of the meter follow exactly the same lines. D is multiplied by 2 or 3, as the case may be, and we form the set of all sounded notes whose durations are greater than the new value of D. If at any stage there is just one such note in this set, then the procedure fails to provide a regular interpretation of the sequence, since we need at least 2 notes of duration greater than D in order to determine D′. But if, for a given value of D, there are no sounded notes of duration greater than D, then the process comes to an end, and the metrical units of duration D can be identified as the bars of a regular passage.

To show how the procedure works in practice, we can apply it to the note sequence (4) of the first section. The shortest notes are the quavers, and these reveal the lowest level of the meter. The sounded notes that exceed a quaver in length are the crotchets and the dotted crotchets; the highest common divisor of their spacings is a crotchet, so that D′/D = 2. The next metrical unit above the quaver is therefore a crotchet unit. The only notes longer than a crotchet are the dotted crotchets, and their spacing is a dotted minim, equal to three crotchets, so that the duration of the next metrical unit is a dotted minim. There are no notes longer than a dotted minim, so the process terminates, and we obtain a rhythmic interpretation of the sequence as the realization of a regular passage in which the value of each bar is a dotted minim, and the meter is [3 2]. This is the interpretation given in (5).

Note sequence (8) of the first section supplies another example. The shortest notes are the quavers, and the notes longer than a quaver are spaced by dotted crotchets, so that the quaver units must be grouped in threes forming dotted crotchet units. There are no notes longer than a dotted crotchet, so the process terminates, yielding an interpretation of the sequence as the realization of a regular passage consisting of dotted crotchet bars as shown in (9). The associated meter is simply [3].

For the isochronous sequence discussed in example (a) of the first section the procedure finds the shortest metrical unit—of value, say, one crotchet—but as there are no sounded notes longer than a crotchet the process immediately halts, implying that every bar contains just one note. Although this interpretation is rather uninteresting, it does at least preclude the implausible interpretations shown in (1), (2), and (3).

As a final example we may consider the note sequence that consists of a single crotchet followed by several quavers. The quaver immediately establishes itself as the shortest metrical unit, but now there is only one note longer than a quaver. The presence of this note indicates that if the passage is regular there must exist metrical units above the quaver level. But we need at least two notes longer than a quaver to delimit this higher metrical unit, and only one note is available. This shortage of evidence corresponds to the fact that the quavers might indeed be grouped in either twos or threes:



In short, there exists a parsing algorithm for note sequences which, when it succeeds, delivers an interpretation of the sequence as the realization of a regular passage. When the algorithm fails at any stage, its failure is evident from the fact that either D′ is not computable (because there is only one sounded note of duration greater than D) or the computed value of D′/D is not a power of 2 or a power of 3.

We can now return, with a clearer insight, to the question why listeners regard some rhythmic interpretations of note sequences as so much more plausible than others. We propose the following answer: that if the parsing algorithm just described succeeds in interpreting a note sequence as the realization of a regular passage, then this interpretation will be implicit in the listener's interpretation of what he hears. We may describe one interpretation as being "implicit" in another if all the metrical units of the former are present in the latter; thus an interpretation that specifies the rhythmic structures of all the bars of a regular passage is implicit in a richer interpretation that goes further and groups the bars together in twos or threes. Thus of the two interpretations shown below, the first is implicit in the second:

We close this section with a pair of real musical examples, to illustrate how the rhythmic ambiguity inherent in the first few notes of a sequence can be resolved by later notes, on the assumption that the sequence is the realization of a regular passage. The examples are taken from the second movement of Beethoven's Piano Sonata Op. 109 and the second subject of the first movement of Brahms's Symphony No. 2. The relative note values (which in the first example disguise the rhythmic structure) are as shown in (17) and (18) below, the first six notes having the same relative values in both passages:


(17)


(18)

For both sequences the parsing algorithm yields a quaver as the shortest metrical unit; the question then arises how the quavers are grouped. In passage (17) the notes longer than a quaver are spaced by the following numbers of quavers: 6,6,3,3, and 3; the highest common divisor of these numbers is 3, so the next metrical unit is of value 3 quavers, equal to one dotted crotchet. There are two notes longer than a dotted crotchet, and these are spaced by two dotted crotchets, so that the highest metrical unit accessible to the algorithm has a value of one dotted minim. In decreasing order of size, the metrical units are therefore a dotted minim, a dotted crotchet and a quaver, and the metre is [2 3]. Parsing the sequence according to this meter we obtain the regular passage shown in (19), which agrees with Beethoven's score.


(19)

In passage (18) the notes longer than a quaver have the following spacings: 6,6,2,2,2,2,2, of which the highest common divisor is 2, so that the quavers must be grouped in pairs, forming crotchets. There are just two notes longer than a crotchet, and these are spaced by three crotchets, so that in descending order the metrical units must be a dotted minim, a crotchet, and a quaver, yielding the meter [3 2]. The sequence is thus interpreted as the realization of a regular passage in $\frac{3}{4}$ time, namely (20):


(20)

It will be noted that our analysis of the Beethoven example differs somewhat from that of Simon (1968), but he also made the point that the full meter of monophonic sequence may not become clear until after a few bars have been heard.

Higher Level Rhythms

Our discussion has so far been confined to monophonic passages in which all the bars are generated by the same standard meter, but the bars are not necessarily grouped together into higher level metrical units. In many passages, however—particularly complete melodies—higher level groupings are musically apparent: In most folk tunes, for example, there is rhythmic structure at all levels, so that the whole passage may be assigned a rhythmic structure with a single root node (Lindblom & Sundberg, 1972). In this section we consider how such higher level rhythms might be perceived by the listener on the basis of the durations of the notes.

The problem of discerning higher level structure arises, in the present context, from the fact that if every bar (except possibly the first) begins with a sounded note, then the bar itself is the highest level metrical unit that is accessible by the parsing algorithm described in the preceding section. The reason is that no note can then have a duration longer than a single bar, so that if D is the duration of a bar, the duration D′ of the units at the next level up cannot be computed from the highest common divisor of the spacings of notes with duration greater than D. For example, if the note durations are as shown in (21)


(21)

only two metrical levels can be discovered by the algorithm, namely the crotchets and the quavers; this *impasse* corresponds to the fact that both (22) and (23)


(22)


(23)

are possible regular interpretations of (21), satisfying the three conditions listed in Section 4, but in these two interpretations the crotchets are grouped in different ways. One does feel, however, that (22) is a more natural interpretation than (23); one therefore looks for perceptual reasons why this might be so.

An obvious property of (21) is that the notes of duration greater than a quaver are spaced by minims ($\frac{1}{2}$ notes), so that $D'/D = 4$, and since this is a power of 2, the algorithm adopts 2D as the duration of the metrical unit at the next level. But it might seem more natural to adopt D' itself, namely 4D, as a higher metrical unit. If this idea is in fact adopted, then one obtains not merely a grouping of the quavers into crotchets but also a grouping of the resulting crotchet units into minim units, with the result shown in (22), and perhaps this is one reason why (22) seems a more natural higher level interpretation than (23). [The resulting rhythmic repetition will undoubtedly reinforce this interpretation, as against (23), but repetition cannot be the whole story because for the note sequence (24) the natural interpretation is still (25) rather than (26).]


(24)


(25)


(26)

We are suggesting, in short, that although higher level rhythms cannot be inferred from the regularity assumption as it stands, the listener may nonetheless perceive such higher level groupings even if the notes of duration greater than D are quite widely spaced, provided that D'/D is still a power of 2 or a power of 3. But we must stress that this conjecture is logically independent of the assumption that the passage is regular in the precise sense defined in the preceding section.

The Role of Phrasing

So far our discussion has been confined to extremely impoverished musical material, in which the only information accessible to the listener is that which can be derived from the times of onset of the notes. In real musical performances, however, the performer will not be content merely to sound the notes at appropriate moments; he will accent some notes more than others, advance or delay the onsets of some of the notes, and play some notes *staccato* and others *legato*. In this section we will sketch a theory of the communicative function of such devices and will suggest that in some cases at least they supply essential clues to the rhythmic structure. In particular we will be concerned with the device of "slurring," although doubtless similar considerations would help one to understand the effects of accent or *rubato*.

In a musical performance of any interest there will be occasional gaps between consecutive notes. If, in a given sequence of notes, the offset of every note coincides with the onset of its successor, then we may describe the sequence as "slurred" or as constituting a single "phrase"; it follows that any sequence whatever can be partitioned into phrases, with a gap between the end of each phrase and the beginning of the next. We propose that phrasing, so defined, is not merely an ornamental device but can serve an essential function in making clear to the listener the rhythmic structure of a passage. The following examples should make clear the basis for this view:


(27)


(28)


(29)


(30)

Thus a sequence of quavers phrased as shown in (27) is likely to be assigned the same meter as the unphrased sequence (28), whereas if the quavers are phrased as shown in (29), the natural rhythmic interpretation will be closely related to that of (30). The effect of the phrasing seems to be to create "virtual" notes of duration equal to the individual phrases; if these durations are sufficiently long they may reveal metrical units of duration greater than the longest notes of the unphrased sequence. To illustrate the role that phrasing can play in the rhythmic clarification of note sequences, we may consider just one simple rule for phrasing the notes of a passage, whether regular or not:

If any metrical unit is divided (into 2 or 3), then slur together all the sounded notes that belong to it except those that descend from its rightmost node.

For a passage in which every bar begins with a sounded note, the effect of this phrasing rule is to replace the sounded note by a virtual note which lasts for at least half the bar, or two-thirds of the bar if the meter is ternary at the highest level. It is not difficult to see, furthermore, that the resulting sequence of virtual notes will be regular even if some of the bars of the unphrased sequence were originally syncopated:

So if a passage is phrased according to the rule in question, then the resulting sequence will be amenable to the parsing algorithm of the fourth section, and it becomes possible for the listener to establish its meter on the basis of the virtual sequence and to use this meter (assuming it does not change from bar to bar) for parsing the individual notes of each phrase and thereby arriving at a rhythmic interpretation in which not necessarily all of the bars are unsyncopated.

To illustrate the use of phrasing for revealing the rhythmic structures of even syncopated passages we may consider a final pair of examples (derived from the opening of Bach's D Minor Klavier Concerto). The note values are as shown in (31),

(31)

and if the passage is slurred according to the given phrasing rule, the result is as shown in (32); the corresponding virtual sequence is given in (33), and leads to the rhythmic interpretation indicated by (34), which is of course the interpretation that Bach intended.

(32)

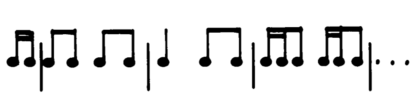
(33)

(34)

If, however, Bach's own bar lines are moved forward by one quaver, then the application of the phrasing rule leads to the slurring given in (35), corresponding to the virtual sequence (36). Applying the parsing algorithm to this sequence we obtain the quite different rhythmic interpretation indicated in (37).

(35)

(36)



(37)

The reader will, perhaps, agree that such an inappropriate phrasing of the passage in question is liable to quite mislead the listener as to its rhythmic structure. The phrasing rule that we have used for illustration is not, of course, intended as an infallible guide to musical performance, but merely an illustration of our thesis that phrasing is not merely an ornamental device but can have a crucial role to play in the elucidation of rhythmic structure.

Conclusions

1. Any given sequence of note values is in principle infinitely ambiguous, but this ambiguity is seldom apparent to the listener.

2. It appears that in choosing a rhythmic interpretation of such a sequence the listener is guided by a strong assumption: that if the sequence can be interpreted as the realization of a regular passage in the sense of the fourth section, then that is how the listener will interpret it.

3. The listener may well group the bars of this passage into higher level metrical units, for reasons of the kind discussed in the fifth section.

4. Phrasing can make an important difference to the way in which the listener perceives the rhythm of a sequence, and even syncopated sequences which would otherwise be difficult to interpret can be rhythmically clarified by slurring the notes appropriately.[1]

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